brainnets: An opensource graph theoretical analysis library and application

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about

- A new, opensource C++ library for graph analysis, tailored for exploring functional brain connectivity.
- Focused on the analysis and visualization of graph features and graph distances.
- Move from statistical analysis to graph analysis.
- Employ well-established graph theoretical methods that retain any topological and structural information.
- Contains all the necessary methods to explore segragation, integration and centrality from both global and local aspects of a graph.

the library

```
// Suppose the following two matrices contain
// some data...
Eigen::MatrixXf mtx1;
Eigen::MatrixXf mtx2;

// Compare matrices
float score =
    brainnets::distances::spectral_k_euclidean(
        mtx1, mtx2, 4
    );

// Compute the MST of the first matrix.
std::tuple<Eigen::MatrixXf, Eigen::MatrixXi> mst =
    brainnets::mst::minimum_spanning_tree(mtx2);
Eigen::MatrixXf filtered_mtx = std::get<0>(mst);

// Compute the Local Efficiency.
float local_eff = brainnets::local_efficiency(mtx);
```

https://github.com/makism/brainnets

MIT License ✔
you can use it to your heart's content

open source ✔
well, we've put it on github; just clone it, modify it and use it

extensive tests ✔
when we introduce new concepts or change something even something small, everything is checked and validate throught unit tests

WIP

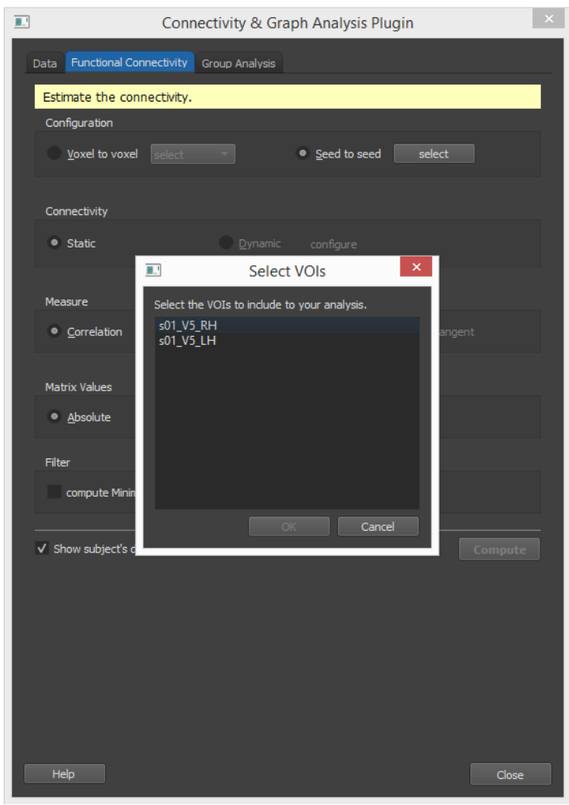
documented methods ✔
provide high quality and scientific documentation

WIP

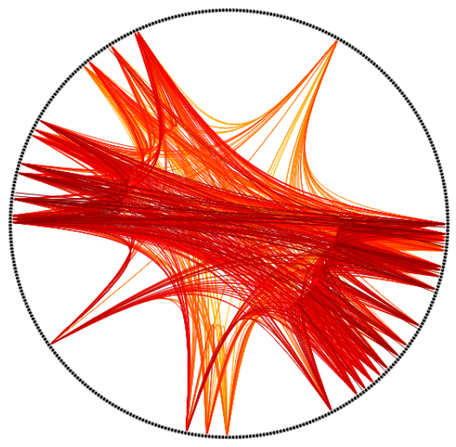
the application

- Integration of the library into BrainVoyager.
- Provide a GUI to utilize all the capabilities of *brainnets*.
- Construct a pipeline to setup a group-analysis.
- Export (pre)processed matrices and visualization results.

Configure the pipeline

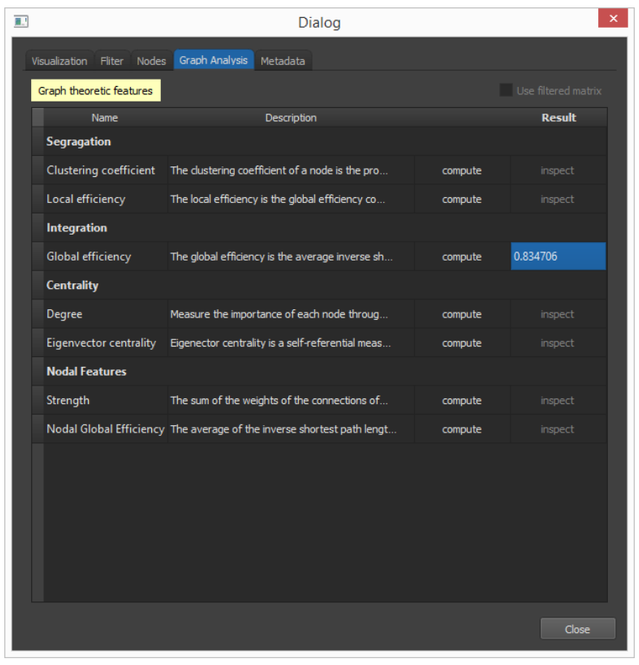


Visualize

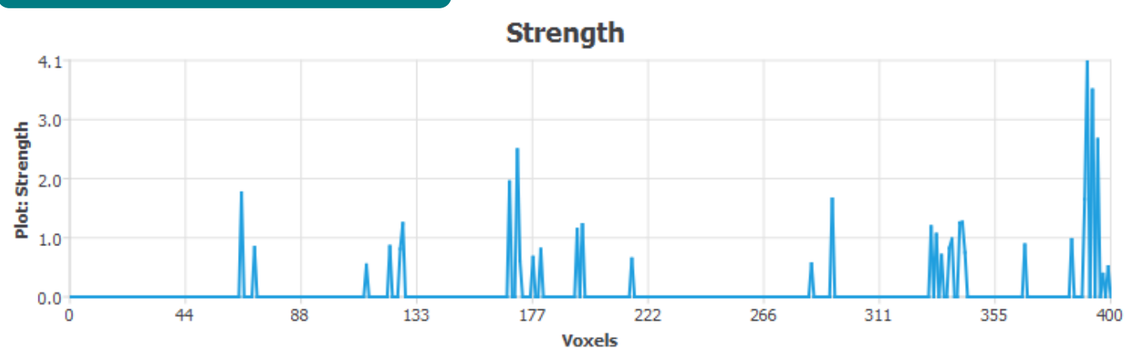


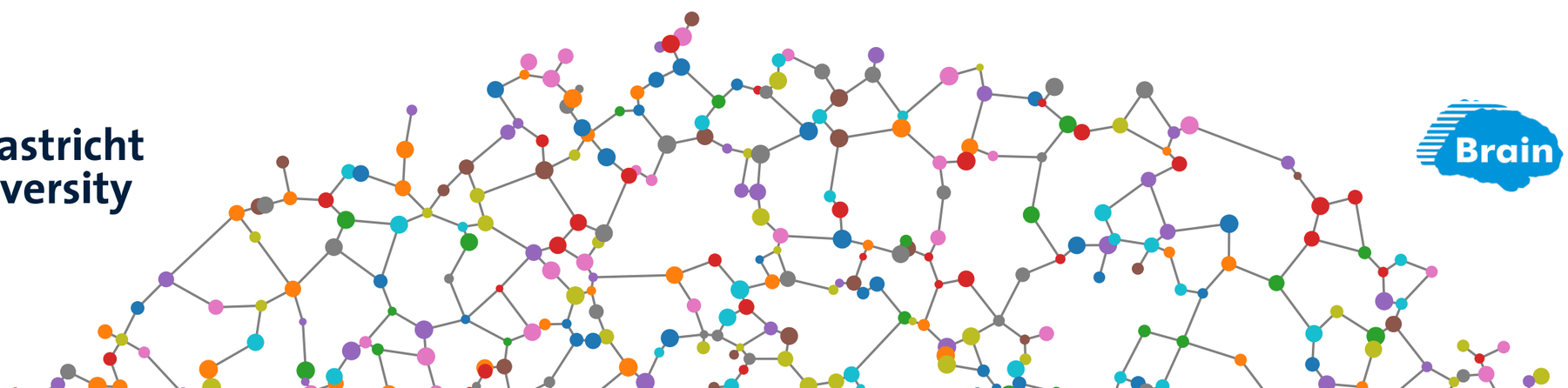
Interactive Circular Plot

Extract features



Plot results



Maastricht University

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